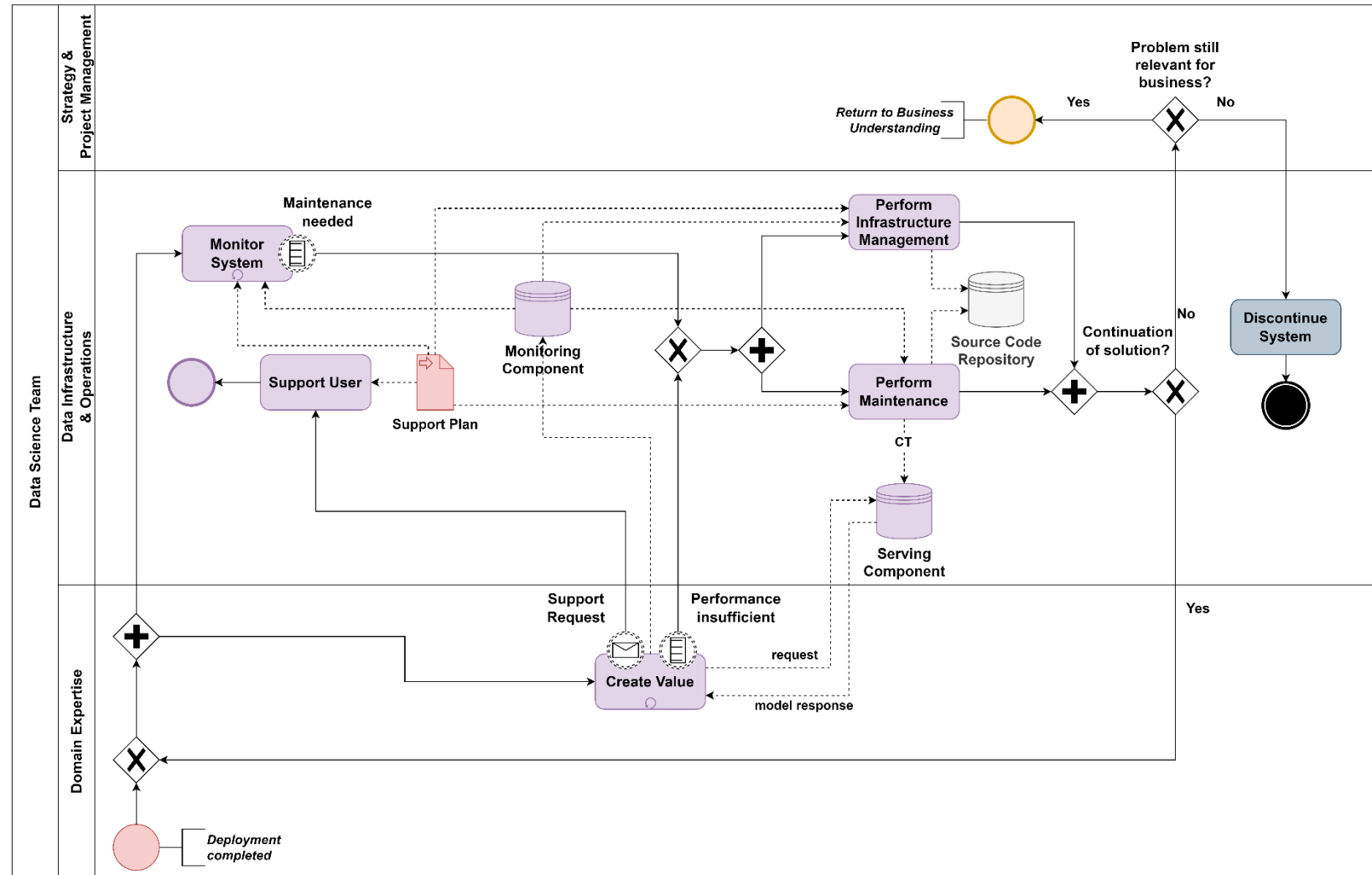


Utilization

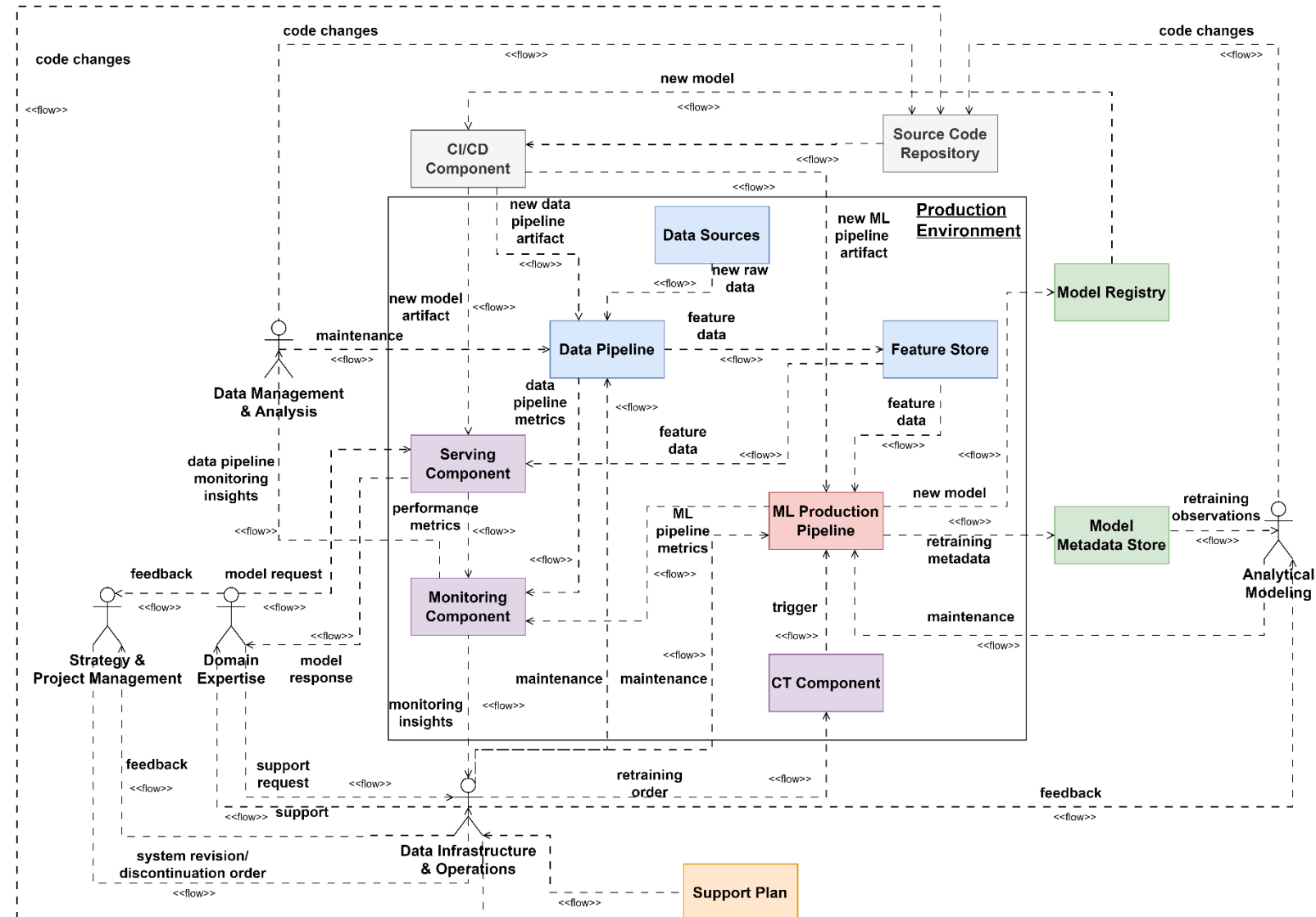
Utilization – Process View

- Use, support, and continuous evaluation of the analytical artifacts
- Main tasks
 - Create Value
 - Support Use
 - Monitor System
 - Perform Maintenance
 - Perform Infrastructure Management
 - (Discontinue System)



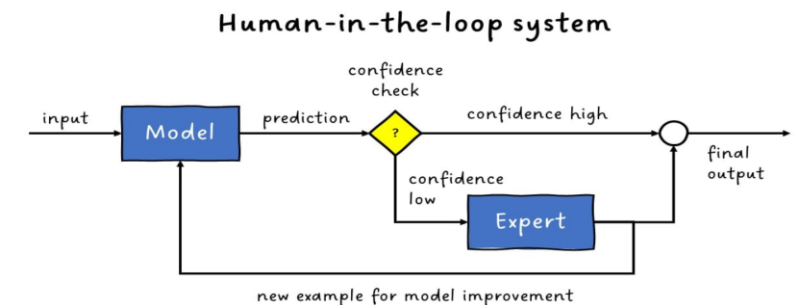
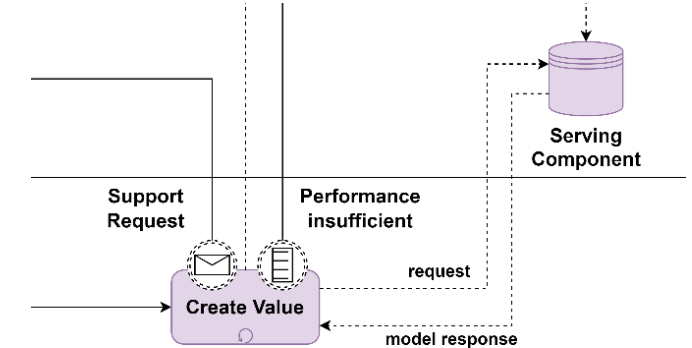
Utilization – Information Flow View

- Utilization is shaped by the business objectives and deployment considerations
 - Users/applications access model through serving component for supporting business processes
 - Monitoring component as the basis for ensuring the system's continued suitability through retraining and infrastructure management



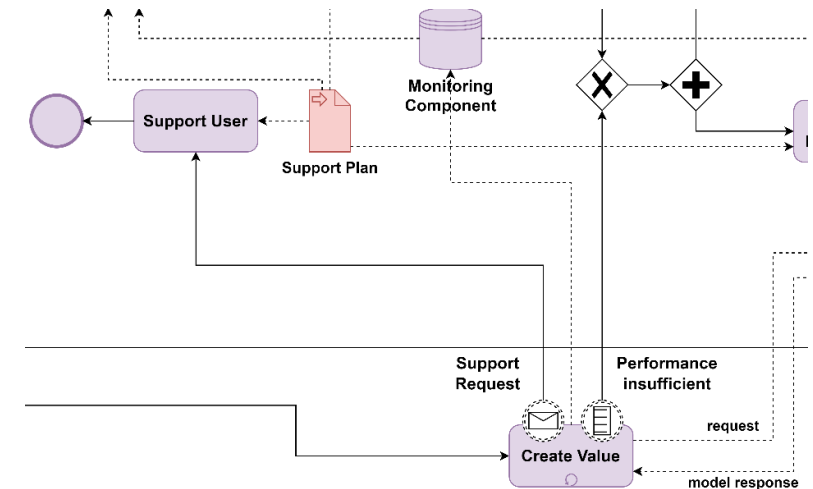
Create Value

- Goal: domain users use analytical artifacts to achieve initially postulated business objectives
- Concrete instantiation depends on product type
- Enabled by user training during the business integration task
- Model governance
 - overall process for how an organization controls access, implements policy, and tracks activity for models and their results
 - How are the model results consumed by downstream systems/processes?
 - Role of explainability
 - Effective model governance crucial for minimizing risk



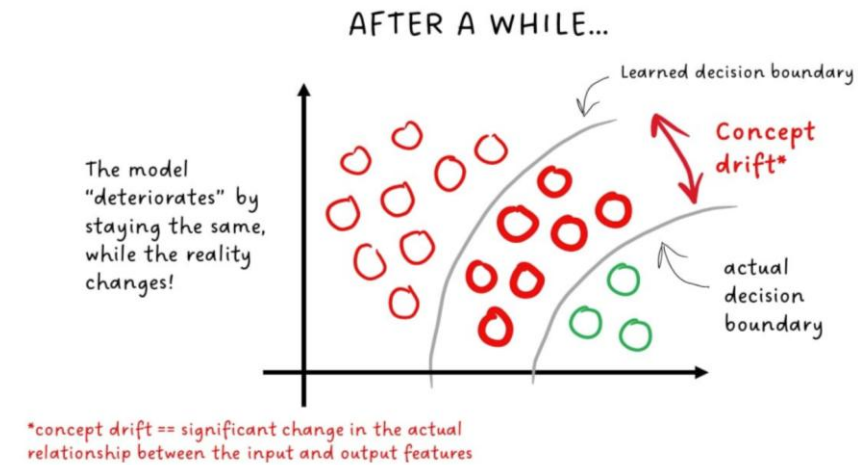
Support User

- Goal: helping users in case of incidents that cannot be solved with available documentation
- Documentation of encountered incidents
 - including potential solutions and workarounds
- Support requests can also be input for maintenance activities



Monitor System

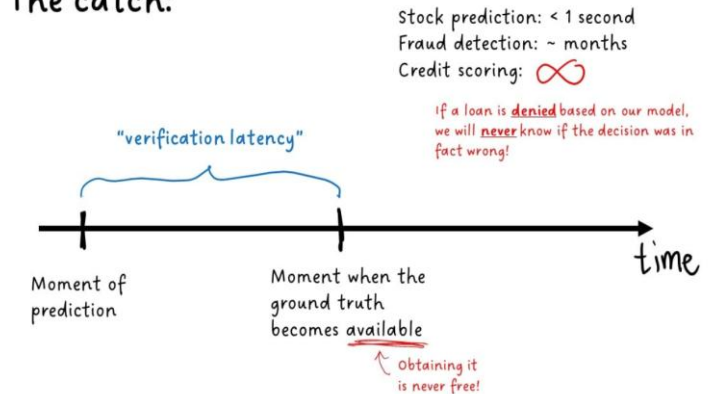
- Goal: continuously evaluate the analytical artifacts to ensure their contribution toward achieving business objectives
 - Based on performance indicators defined in the support plan
- ML models can become stale over time and degrade in performance
 - Other problems: Code maintenance, scaling issues, etc.
- Types of monitoring
 - Statistical monitoring: focuses on the input and output data, including predictions
 - Computational monitoring: focuses on technical metrics
- typically supported by multiple tools and dashboards



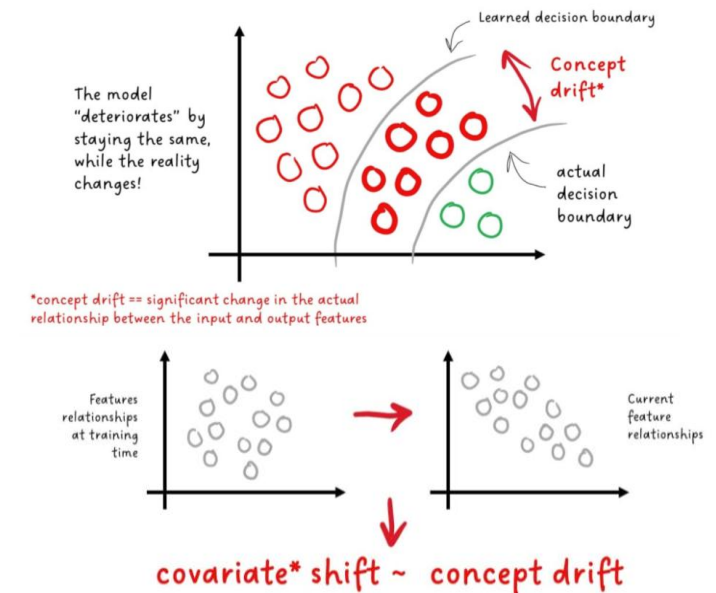
Monitor System

- Challenges of model monitoring
 - Requires storing requests, model predictions, and ground truth
 - Difficulty of capturing ground truth data
 - Availability of ground truth data → verification latency
 - time and effort for procurement
 - Focus on data/feature monitoring
- Drift
 - Concept drift: changes in relationship between feature and targets
 - Difficult to detect
 - Often affects the business impact of the model
 - Data drift/covariate shift: shifts in the input data distribution
 - Possible to detect using statistical methods
 - Not every drift impacts performance

The catch:



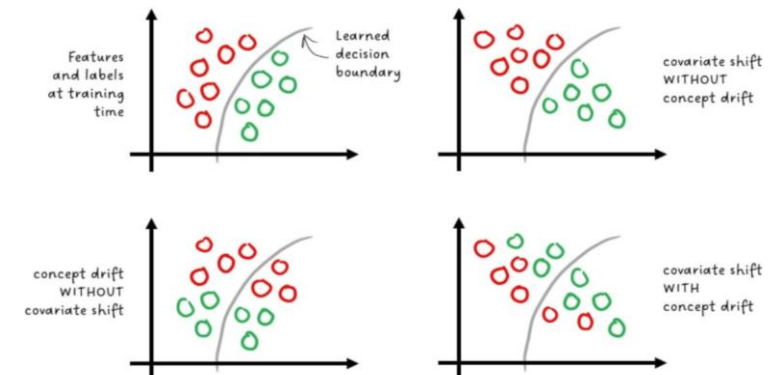
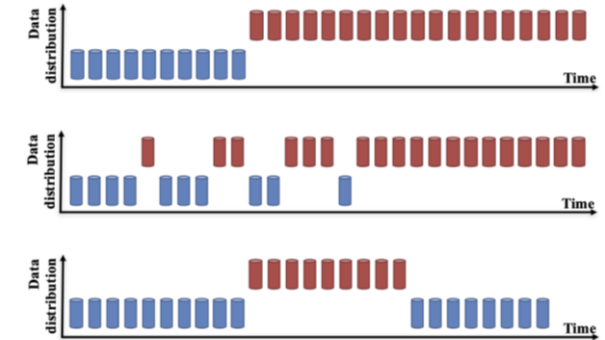
AFTER A WHILE...



* features == "covariates" in statistical jargon

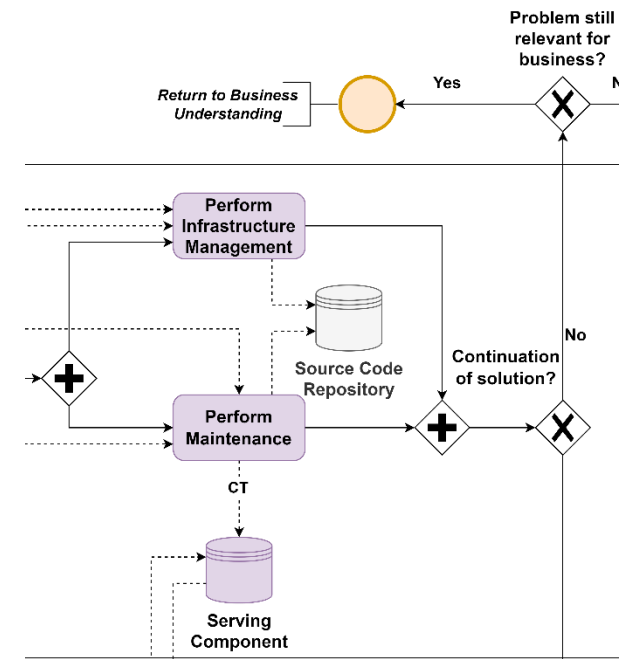
Monitor System

- Reasons for drift
 - External events: viral trends, policy changes
 - Unmodeled seasonality: demand forecasting seasonal events
 - Changes in data-generation process: new update to the data collection
 - Evolving user behavior: habits, patterns, preferences are constantly changing
 - ...
- Dynamics of drift: sudden, gradual, seasonal
- Effect of covariate shift on concept drift: unknown, no effect, negative, positive
- Covariate shift detection: univariate, multivariate, continuous methods (Hellinger, Jensen–Shannon, Wasserstein)



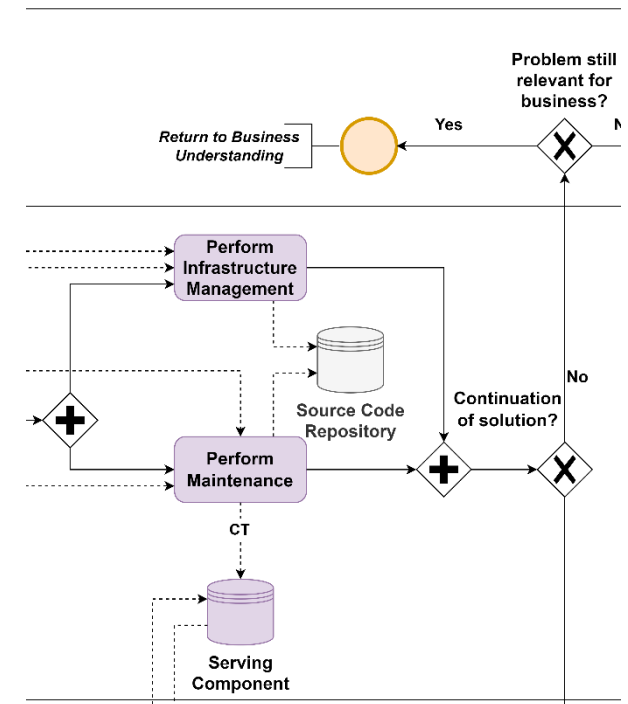
Perform Maintenance

- Goal: improve deployed systems based on thresholds and measures defined in the support plan
 - relies on observations captured by the monitoring component
- Root cause analysis
 - Covariate shift
 - Concept shift
- Issue resolution strategies
 - Retraining: keep model up-to-date with recent patterns
 - Requires additional data and computation
 - Facilitated by ML pipeline and CI/CD component
 - model improvement: new models, new derived features
 - dataset improvement: clean, enrich, enlarge dataset
 - Considerations of costs → approach?
 - Refactoring use case
 - Changing downstream processes around model predictions



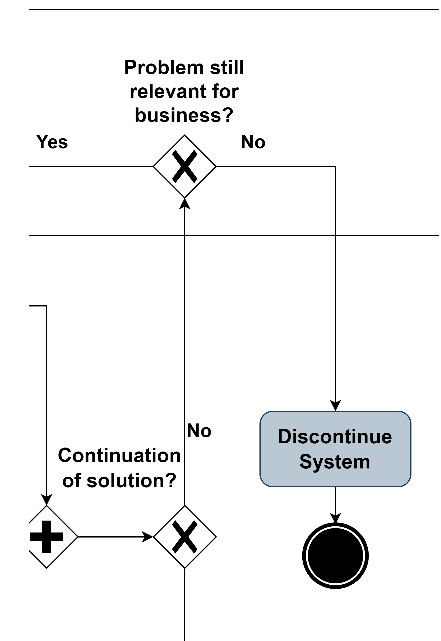
Perform Infrastructure Management

- Goal: ensure that the solution sustains non-functional and operational requirements
 - Application performance
 - latency
 - ...
- Maintenance activities specifically concerning technical supporting IT infrastructure
 - Model training and serving are enabled by the appropriate computational resources (Model Training Infrastructure)
 - computational monitoring metrics as indicators
- Impacted by the request volume and data requirements (e.g., retraining)
 - Scalability
 - Optimization
 - ...



Discontinue System

- In case of a decision against the continuation of the solution:
 - Rework: return to Business Understanding
 - Termination: full discontinuation of the analytical system in its current form (end of the DS lifecycle)
- Reasons
 - Ineffectiveness of maintenance measures
 - Low return on investment
 - Irrelevance of the originally addressed business problem
 - Lacking user acceptance
 - ...



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